

Xylem Tissue

Subject & topics: Biology | Physiology | Exchange & Transport Stage & difficulty: A Level P2

Part A Function

Xylem tissue is used to transport from the to the .

It also provides plants with .

Items:

- carbon dioxide
- stems and leaves
- structural support
- water (and minerals)
- roots
- oxygen
- sugars
- light

Part B Structure

Xylem tissue includes xylem vessels (the main structures that transport) , parenchyma cells (which can store water, nutrients, and tannins), as well as some other cell types and structures.

Xylem vessels are initially formed as columns of cells. These cells produce a strong polymer called , which is deposited in the cell walls. The cells then die, leaving behind hollow tubes surrounded by the polymer.

Because the polymer is very strong and rigid, xylem vessels also provide the plant with .

Items:

- starch
- carbon dioxide
- keratin
- water
- structural support
- chitin
- lignin
- light

Part C

Transport mechanisms

Various mechanisms are involved in the movement of water into and through a plant. Match the mechanism to the description in the table below.

Mechanism	Description
	the evaporation of water out of the leaves (through stomata) which pulls water towards the leaves
	the attraction between water molecules that ensures that they are pulled upwards together
	the movement of water molecules from a region of higher water potential to a region of lower water potential through a partially permeable membrane (e.g. from the soil into the root hair cells)

Items:

- cohesion
- transpiration
- translocation
- osmosis
- active transport

Part D

Xylem statements

Which of the following statements are correct? Select all that apply.

- ☐ xylem is found in the roots, stems, and leaves of a plant
- ☐ xylem vessels are non-living, but other parts of xylem tissue (e.g. parenchyma cells) are living
- ☐ xylem tissue carries out photosynthesis
- ☐ xylem is only found in the main stem of a plant
- ☐ xylem transports sugars from the leaves to the non-photosynthetic parts of the plant (e.g. the roots)
- ☐ xylem transports water from the roots to the stems and leaves
- ☐ xylem tissue is entirely non-living
- ☐ xylem provides structural support to plants

Phloem Tissue

Subject & topics: Biology | Physiology | Exchange & Transport **Stage & difficulty:** A Level P2

Part A Function

Phloem tissue is used to transport from the to the .

This process is called .

Items:

- transpiration
- translocation
- water
- photosynthetic parts of the plant (e.g. leaves)
- non-photosynthetic parts of the plant
- photosynthates (e.g. sugars and amino acids)

Part B Structure

The main vessels in phloem tissue formed by cells called , which form a column. The cell walls between these cells form large holes and are referred to as , which allows material to move between the cells easily. These cells also lose their nucleus, tonoplast, and some other organelles.

The vessels are surrounded by cells called , which are connected to the vessel cells by . These cells provide energy and materials to the vessel cells.

Items:

- gap junctions
- sieve plates
- sieve tube elements
- companion cells
- lignin
- stomata
- plasmodesmata

The mass flow hypothesis

The mass flow hypothesis offers an explanation of how organic compounds (e.g. sugars) are translocated via phloem tissue.

The hypothesis states that in areas (e.g. leaves, where sugars are produced during photosynthesis), water potential is as the concentration of soluble sugars increases. This causes water to flow into these areas, increasing the hydrostatic pressure.

In areas (e.g. growing roots, where sugars are being used up during respiration), water potential is as the concentration of soluble sugars decreases. This causes water to flow out of these areas, decreasing the hydrostatic pressure.

The water will then flow through from the high pressure area to the low pressure area, taking soluble sugars with it.

Items:

- increased
- decreased
- sink
- source
- sieve tube elements
- xylem vessels

Xylem vs Phloem

Subject & topics: Biology | Physiology | Exchange & Transport

Stage & difficulty: A Level P1

Part A

Tissue comparisons

In the table below, identify whether each statement applies to xylem, phloem, or both.

Statement	Tissue
found in roots, stems, and leaves	
responsible for translocation	
responsible for transpiration	
contains living cells	
main vessels are composed of lignified non-living tissue	
main vessels are composed of sieve tube elements	

Items:

- xylem
- phloem
- both

Part B

Vascular bundles

Xylem and phloem tissue are found together in "vascular bundles".

A student was investigating the vascular system in a flowering plant and observed a vascular bundle. They identified features of a mature sieve tube element.

The cross-sectional area of a typical xylem vessel is six times greater than the cross-sectional area of this mature sieve tube element.

The cross-sectional area of a typical xylem vessel is

What is the cross-sectional diameter of this sieve tube element? Give your answer to the nearest integer.

Assume that both cross-sectional areas are circular.

Part C

Radioactive tomatoes

Tomato plants were placed in sunlight in an atmosphere containing radioactively labelled carbon dioxide. After a period of time, radioactively labelled sugars were detected in the roots of the plants.

Drag the statements below into the correct chronological order on the right (with the first step at the top and the last step at the bottom). Only some of the statements will be needed.

Available items

- carbon dioxide diffused into the roots
- carbon dioxide entered the leaves
- the carbon dioxide was transported to the roots by translocation
- the carbon dioxide was transported to the roots by transpiration
- the carbon dioxide was converted into sugars in the roots
- the carbon dioxide was converted into sugars in the leaves
- the sugars were transported to the roots by translocation
- the sugars were transported to the roots by transpiration

Question deck:

STEM SMART Biology Week 22 - Plant Nutrient & Water Transport

Stomata: Structure & Function

Subject & topics: Biology | Physiology | Exchange & Transport

Stage & difficulty: A Level P1

Stomata (singular: stoma) are pore structures found mostly on the underside of leaves. Their main function is to allow gas exchange between the inside of the leaf and the surrounding air, though they also play an important role in transpiration.

Figure 1 shows an open stoma on the left and a closed stoma on the right.

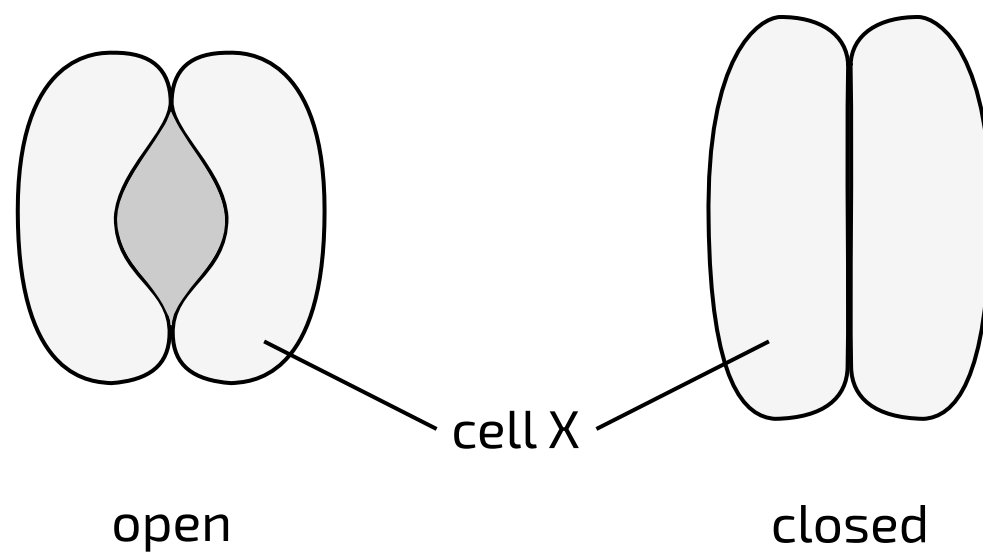


Figure 1: An open stoma (left) and a closed stoma (right).

Part A

Cell X

What type of cell is cell X in **Figure 1**?

Part B

Diffusion directions

Fill in the blanks below to describe the diffusion of gases between a leaf and the surrounding air when leaf is in direct sunlight and the stomata are open.

- Oxygen will diffuse the leaf.
- Carbon dioxide will diffuse the leaf.
- Water vapour will diffuse the leaf.

Items:

into

out of

Part C

Opening stomata

In many plants, stomata are opened in response to high light intensity. ATP generated in the light-dependent stage of photosynthesis is used to actively transport ions and other solutes into the guard cells.

Drag the items below into the correct order on the right to show how this causes stomata to open.

Note that not all of the items below are part of the correct sequence of events, and so you should not use all of the items below.

Available items

guard cells become **turgid**

guard cells become **flaccid**

the water potential inside guard cells **increases** as a result of solute movement

the water potential inside guard cells **decreases** as a result of solute movement

water moves **into** guard cells by osmosis

water moves **out of** guard cells by osmosis

the stomatal pores **open**

Part D

Closing stomata

In many plants, stomata are closed in response to low water levels. Absciscic acid (ABA) is released by root cells and travels through the xylem to the leaves. In the leaves, ABA binds to receptor proteins, resulting in ions and other solutes being exported out of the guard cells.

Drag the items below into the correct order on the right to show how this causes stomata to close.

Note that not all of the items below are part of the correct sequence of events, and so you should not use all of the items below.

Available items

guard cells become **turgid**

guard cells become **flaccid**

the water potential inside guard cells **increases** as a result of solute movement

the water potential inside guard cells **decreases** as a result of solute movement

water moves **into** guard cells by osmosis

water moves **out of** guard cells by osmosis

the stomatal pores **close**

Plant Water Loss

Subject & topics: Biology | Physiology | Exchange & Transport

Stage & difficulty: A Level P2

Part A

Stomata control

Which of the following statements correctly explain how the stomata are controlled? Select all that apply.

- ☐ when water levels are **high**, the guard cells become turgid, which causes each stoma to **open**
- ☐ when water levels are **high**, the guard cells become turgid, which causes each stoma to **close**
- ☐ when water levels are **low**, the guard cells become less turgid, which causes each stoma to **open**
- ☐ when water levels are **low**, the guard cells become less turgid, which causes each stoma to **close**

Part B

Low water environments

What is the general name given to plants that are adapted to living in low water environments?

Which of the following adaptations are commonly found in these plants? Select all that apply.

- ☐ thick waxy cuticle
- ☐ very thin cuticle or no cuticle
- ☐ many stomata
- ☐ few stomata
- ☐ leaves with a large surface-area-to-volume ratio
- ☐ leaves with a small surface-area-to-volume ratio
- ☐ large root system
- ☐ small root system

Part C

High water environments

What is the general name given to plants that are adapted to living in high water environments?

Which of the following adaptations are commonly found in these plants? Select all that apply.

- ☐ thick waxy cuticle
- ☐ very thin cuticle or no cuticle
- ☐ many stomata
- ☐ few stomata
- ☐ leaves with a large surface-area-to-volume ratio
- ☐ leaves with a small surface-area-to-volume ratio
- ☐ large root system
- ☐ small root system

Capillary Action and Water Transport

Subject & topics: Biology | Physiology | Exchange & Transport

Stage & difficulty: A Level C1

An experiment was performed to investigate whether capillary action could account for the movement of water from roots to leaves in a plant.

Figure 1 shows apparatus used to model this effect using narrow glass capillary tubing. The inner diameter of the capillary tubing is 0.5 mm.

The initial height of the water in the tubing was recorded as 0.2 cm.

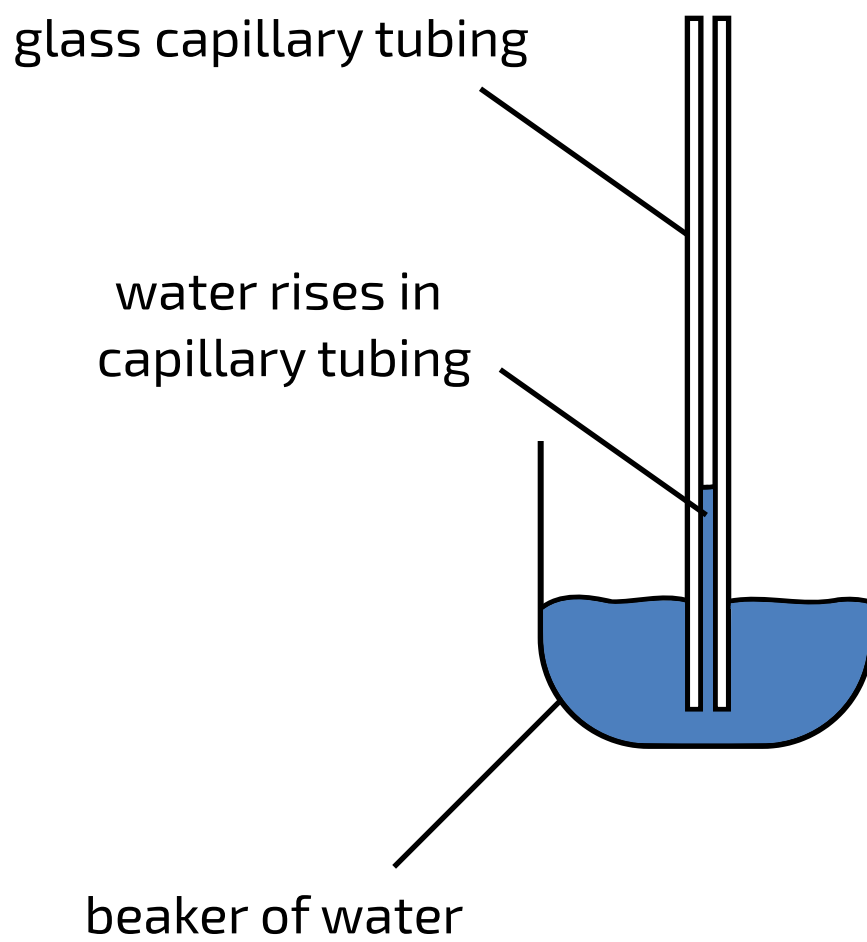


Figure 1: Capillary action experiment. A glass capillary tubing was placed in a beaker of water, and the initial height of water in the capillary tubing was recorded. Over time, the water in the capillary tubing rises as a result of capillary action.

Part A

Rate of transport

After 2 minutes the height of the water was recorded as 1.8 cm.

Calculate the rate of water movement in $\text{mm}^3 \text{min}^{-1}$. Give your answer to 3 significant figures.

Part B

Transpiration processes

In the experiment above, the water would only travel upwards a certain distance and then stop. In plants, water continuously travels upwards.

This suggests that capillary action cannot fully account for water transport in plants.

Which of the following additional processes are occurring in plants but **not** in the experiment above?

- ☐ adhesion between the water molecules and the surface of the tube (xylem/capillary tubing)
- ☐ evaporation of water, pulling the water upwards
- ☐ movement of water into the system
- ☐ cohesion between water molecules, allowing the water to be pulled upwards

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Question deck:

STEM SMART Biology Week 22 - Plant Nutrient & Water Transport

Water Usage Analysis

Subject & topics: Biology | Physiology | Exchange & Transport **Stage & difficulty:** A Level C2

Water molecules can contain different isotopes of H and O. The "light" isotopes (^1H and ^{16}O) are by far the most abundant. The "heavy" isotopes (^2H and ^{18}O) are present at much lower levels.

The ratio of different stable isotopes ($^2\text{H}/^1\text{H}$ or $^{18}\text{O}/^{16}\text{O}$) in water samples extracted from plant tissues can be used to track water usage by plants.

Part A

Isotope analysis

To determine whether a particular water sample is enriched or depleted of heavy isotopes, that sample has to be compared to a reference standard. The most widely used standard is the Standard Mean Ocean Water (SMOW); the isotope ratio in a distilled sample of ocean water.

In SMOW there is approximately 1 ^2H atom in every 6421 H atoms. Calculate $^2\text{H}/^1\text{H}$. Give your answer to 3 sf.

Part B

Isotope abundance

Relative stable isotope abundances are expressed in parts per thousand (‰) compared to the SMOW reference. For example, relative isotope abundance of ^2H (denoted $\delta^2\text{H}$) is calculated as:

$$\delta^2\text{H} = \frac{^2\text{H}/^1\text{H}_{\text{sample}}}{^2\text{H}/^1\text{H}_{\text{SMOW}}} - 1 \times 1000 \text{ ‰}$$

$\delta^2\text{H}$ differs between water samples because heavy isotopes evaporate more slowly than light isotopes.

Which of the following statements are true? Select all that apply.

- ☐ Water derived from summer rain has lower $\delta^2\text{H}$ and $\delta^{18}\text{O}$ than winter rain.
- ☐ Water derived from summer rain has higher $\delta^2\text{H}$ and $\delta^{18}\text{O}$ than winter rain.
- ☐ In the summer, groundwater has higher $\delta^2\text{H}$ and $\delta^{18}\text{O}$ than water in the upper soil layers.
- ☐ In the summer, groundwater has lower $\delta^2\text{H}$ and $\delta^{18}\text{O}$ than water in the upper soil layers.

Part C

Calculating $\delta^2\text{H}$

A sample of water extracted from plant xylem had a $^2\text{H}/^1\text{H}$ of 1.51×10^{-4} .

Calculate the $\delta^2\text{H}$ for this sample. Give your answer to 2 sf.

Part D

Water usage

A researcher studying water usage in a drought-hit area analysed the $\delta^2\text{H}$ in the xylem sap of a large maple tree, and in smaller shrubs growing at different distances from the tree. Their results are tabulated below. The value at 0 m distance is the xylem sap of the maple tree itself.

Distance from maple tree (m)	0	1	2	3	4	5
$\delta^2\text{H}$ of xylem sap (‰)	−50	−48	−40	−28	−25	−24

The $\delta^2\text{H}$ of rainwater that fell before the drought began was −23 ‰.

Which of the following is a correct interpretation of these data?

- ☐ Plants growing further from the tree took up more groundwater.
- ☐ None of the plants took up groundwater.
- ☐ Plants growing further from the tree took up less groundwater.
- ☐ All plants mostly took up groundwater.

Part E

Water transport

The maple tree has long roots that reach into the groundwater, whereas the neighbouring shrubs have much shorter roots (Figure 1). The water potential in the maple tree roots, ground water and surrounding soil were measured over a three day period (Figure 2).

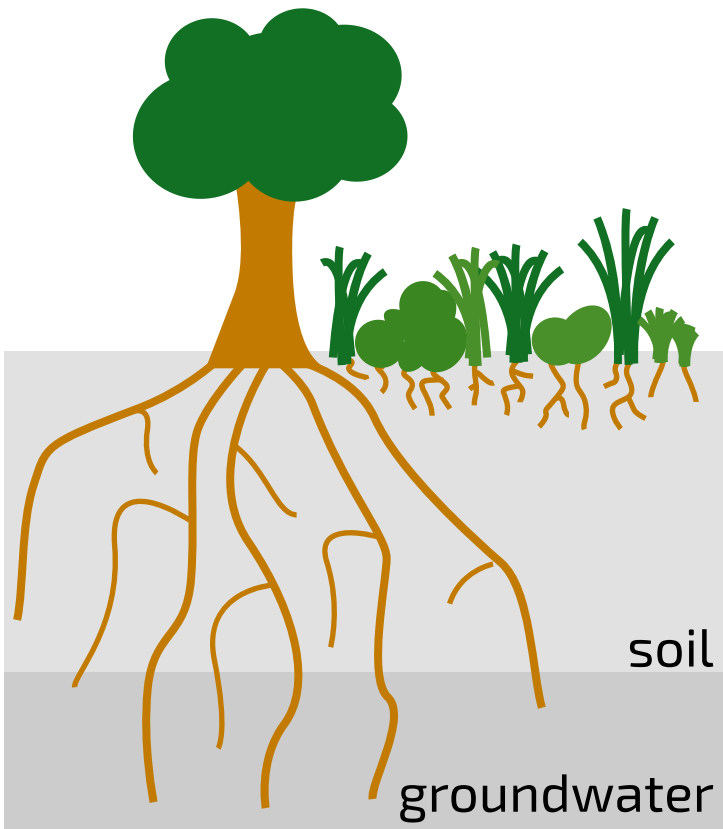


Figure 1: Root architecture of the maple tree and surrounding shrubs.

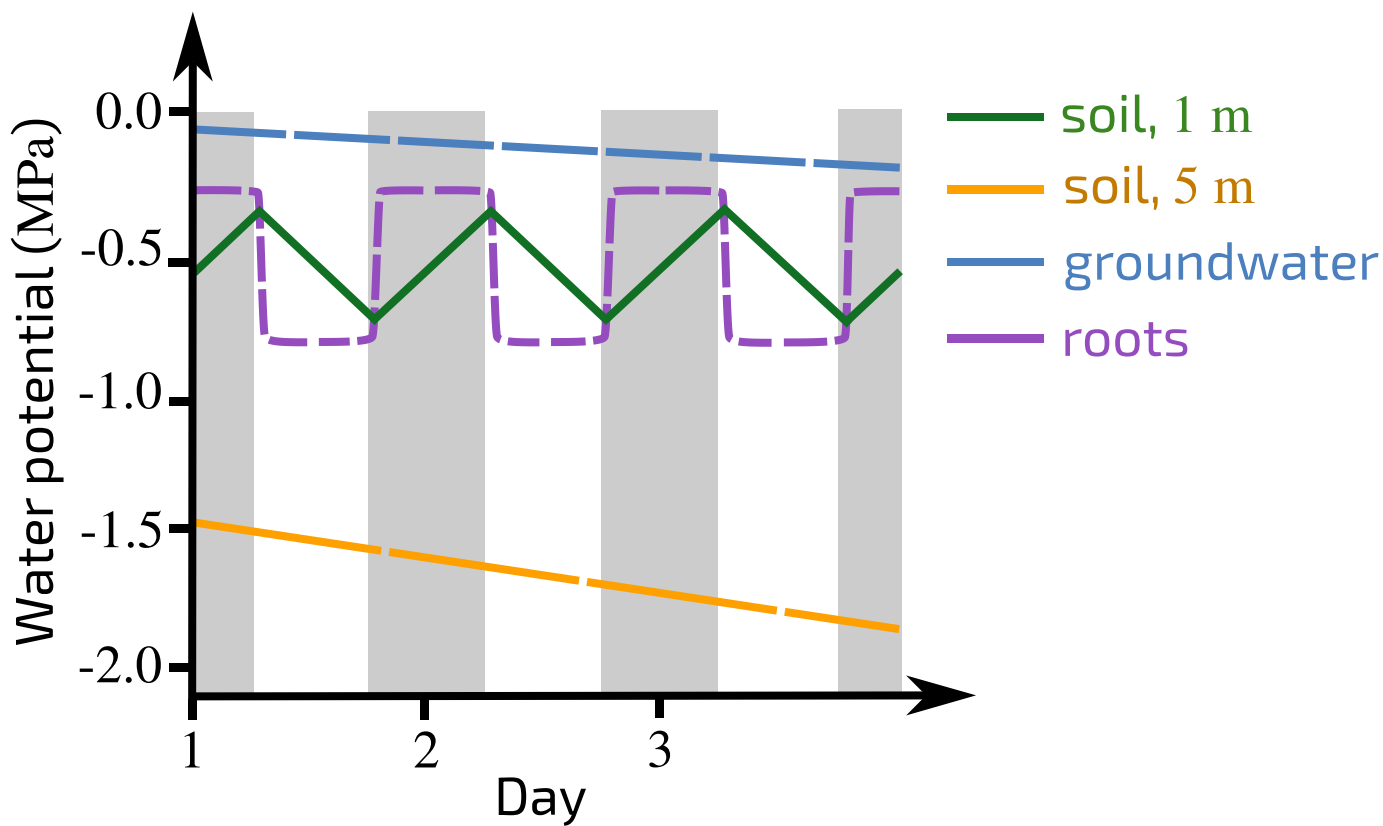


Figure 2: Changes in water potential in and around the maple tree over three days. Measurements were taken from the root xylem, groundwater and from soil samples collected at 1 m (green) and 5 m (yellow) horizontal distance from the tree. Day/night cycles are indicated by light/dark shading.

Using this information, to explain how water moves between the tree root and soil during a drought.

At all times, the water potential in the groundwater is than in the roots, so water moves from the to the .

At night, there is minimal taking place, so water potential in the roots , becoming than the surrounding soil. Therefore, water moves from the to the .

During the day, high levels of take place, so the water potential in the roots , becoming than the surrounding soil. Therefore, water moves from the to the .

Items:

- transpiration
- respiration
- roots
- soil
- groundwater
- increases
- decreases
- higher
- lower